

Scalability & Future Plan

Date	30 June 2026
Team ID	
Project Name	Credit Card Approval
Maximum Marks	1 Mark

This document outlines how the current Smart Card Approve solution can be scaled to support a larger volume of applicants, increased traffic during promotional campaigns, and advanced AI capabilities. It also presents the roadmap for system enhancements, automated risk assessments, and cloud deployment strategies.

Current System Limitations

#	Limitation	Impact	Priority
1	MySQL database hosted on a local server	May experience slower performance with a large number of concurrent users.	Medium
2	Machine Learning model is trained offline	Credit risk assessment accuracy depends heavily on manual, periodic retraining with updated financial	Medium
3	Manual document verification	Supporting documents are not automatically verified, increasing processing time.	High

Scalability Plan

#	Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local server	Deploy the web application on cloud platforms (AWS, Azure, or Render) utilizing load balancers to handle peak application traffic.
2	Data Storage	Local MySQL database	Migrate to cloud-managed MySQL or

			PostgreSQL with automated backup and replication.
3	Performance	Synchronous request processing	Implement asynchronous task queues (Celery + Redis) to process heavy machine learning prediction and risk-scoring requests efficiently.
4	Security	Basic username/password authentication	Implement HTTPS, JWT authentication, role-based access control (RBAC) for bank underwriters, and multi-factor authentication (MFA).

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Deploy Smart Lender on cloud using Docker containers and CI/CD pipeline	Sep 2026	Enables secure, highly available online access and seamlessly supports thousands of concurrent applicants.
Phase 3	Integrate real-time Credit Bureau APIs (e.g., CIBIL) and document verification services	Nov 2026	Improves prediction accuracy and reduces manual verification efforts
Phase 4	Implement automated MLOps pipeline for continuous model retraining using new loan datasets	Feb 2027	Keeps the approval/risk model highly accurate against changing macroeconomic conditions and evolving consumer trends.
Phase 5	Develop Android/iOS mobile applications with AI-powered financial recommendations and multilingual support	May 2027	Enhances user accessibility, boosts digital credit card acquisition rates, and offers personalized financial insights.